WORKED SOLUTION

STEP III 1993, Question 1

Pure • Arc Length and Volume

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q1

The key idea

The double-angle formula $\cos 2\theta = 1 - 2\sin^2\theta$ turns the parametric curve straight into a Cartesian one, and the range of θ tells us exactly which piece of the parabola we get. For the length, $\sqrt{1+16x^2}$ is the classic signal for a hyperbolic substitution, because $1 + \sinh^2 u = \cosh^2 u$. The volume is an ordinary volume of revolution once we notice that the ends of P sit on the axis of rotation.

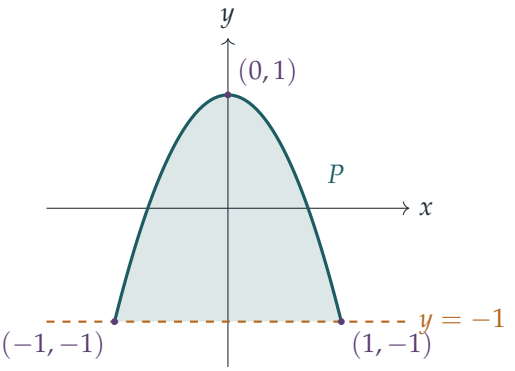
Solution

The curve and its sketch

Since $\cos 2\theta = 1 - 2\sin^2\theta$ and $x = \sin\theta$,

$$y = 1 - 2x^2.$$

So every point of P lies on the parabola $y = 1 - 2x^2$. As θ runs from $-\pi/2$ to $\pi/2$, $x = \sin\theta$ increases steadily from -1 to 1, taking each value exactly once. So P is precisely the arc of the parabola with $-1 \le x \le 1$. It runs from $(-1,-1)$ up to its vertex $(0,1)$ and back down to $(1,-1)$, and it is symmetric about the y-axis.



The length of P

Working in Cartesian form, $\frac{\mathrm{d}y}{\mathrm{d}x} = -4x$, so

$$\text{length} = \int_{-1}^{1} \sqrt{1+16x^2}\,\mathrm{d}x = 2\int_{0}^{1} \sqrt{1+16x^2}\,\mathrm{d}x,$$

using the fact that the integrand is even. (The parametric formula $\int \sqrt{\dot{x}^2+\dot{y}^2}\,\mathrm{d}\theta$ gives the same integral after substituting $x = \sin\theta$.)

? **Ask yourself** Which substitution makes $\sqrt{1+16x^2}$ collapse?

We want $16x^2$ to be a $\sinh^2$, so put $4x = \sinh u$. Then $4\,\mathrm{d}x = \cosh u\,\mathrm{d}u$ and $\sqrt{1+16x^2} = \cosh u$ (positive, since $\cosh u > 0$). The limits $x = 0, 1$ become $u = 0$ and $u = \alpha$, where $\alpha = \sinh^{-1} 4$. Note that $\sinh\alpha = 4$ and $\cosh\alpha = \sqrt{1+16} = \sqrt{17}$. Using $\cosh^2 u = \frac{1}{2}(1+\cosh 2u)$,

$$\begin{aligned} 2\int_0^1 \sqrt{1+16x^2}\,\mathrm{d}x &= 2\int_0^\alpha \cosh u \cdot \tfrac{1}{4}\cosh u\,\mathrm{d}u = \tfrac{1}{4}\int_0^\alpha (1+\cosh 2u)\,\mathrm{d}u \\ &= \tfrac{1}{4}\Big[u + \tfrac{1}{2}\sinh 2u\Big]_0^\alpha = \tfrac{1}{4}(\alpha + \sinh\alpha\cosh\alpha) \\ &= \tfrac{1}{4}(\sinh^{-1}4 + 4\sqrt{17}). \end{aligned}$$

So the length of P is $\sqrt{17} + \frac{1}{4}\sinh^{-1} 4$, as required.

The volume

Both ends of P lie on the line $y = -1$, so when P is rotated about that line it sweeps out a closed surface. The solid it encloses is exactly what we get by rotating the shaded region between P and $y = -1$. A slice at position x is a disc of radius

$$y - (-1) = 2 - 2x^2 = 2(1-x^2)\ (\geq 0 \text{ for } |x| \leq 1).$$

Hence

$$\begin{aligned} V &= \pi\int_{-1}^{1} 4(1-x^2)^2\,\mathrm{d}x = 8\pi\int_0^1 (1-2x^2+x^4)\,\mathrm{d}x \\ &= 8\pi\Big(1 - \tfrac{2}{3} + \tfrac{1}{5}\Big) = 8\pi\cdot\tfrac{8}{15} = \frac{64\pi}{15}. \end{aligned}$$

Checks

- At $\theta = 0$ we get $(0,1)$ and at $\theta = \pm\pi/2$ we get $(\pm 1, -1)$, which agree with $y = 1 - 2x^2$.
- The length $\sqrt{17} + \frac{1}{4}\sinh^{-1} 4 \approx 4.6468$ should be a little more than the length of the two straight chords from $(\pm 1, -1)$ to $(0,1)$, which is $2\sqrt{5} \approx 4.472$. It is. A midpoint-rule evaluation of $\int \sqrt{\cos^2\theta + 4\sin^2 2\theta}\,\mathrm{d}\theta$ over $[-\pi/2, \pi/2]$ also gives 4.64678.

- Pappus's theorem: the region has area $\int_{-1}^{1}(2-2x^2)\,\mathrm{d}x = \frac{8}{3}$, and its centroid is $\frac{4}{5}$ above the axis. So $V = 2\pi \cdot \frac{4}{5} \cdot \frac{8}{3} = \frac{64\pi}{15} \approx 13.404$, which also matches numerical integration.

Common slips

- Rotating about the x-axis by mistake and using radius y instead of $y+1$.
- Forgetting the factor $\frac{1}{4}$ from $\mathrm{d}x = \frac{1}{4}\cosh u\,\mathrm{d}u$, or leaving $\sinh 2\alpha$ unsimplified instead of writing it as $2\sinh\alpha\cosh\alpha = 8\sqrt{17}$.

Beyond STEP

Arc length is a first glimpse of the differential geometry of curves. If we measure position along a curve by arc length s, the speed is always 1, and how fast the direction turns with s is the *curvature*. For the parabola $y = 1-2x^2$ this is largest at the vertex. Hyperbolic substitutions like the one above appear all over first-year calculus. Siklos, *Advanced Problems in Mathematics*, has many more questions in this style.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.